



Release: 12.12.

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Exercise Sheet 9

Exercise 1 — *Wave equation in continuum*

[moderate]

Consider a mono-atomic chain with N atoms of mass M . Show that the N coupled equations of motion

$$M \frac{d^2 u_n}{dt^2} = \sum_p C_p (u_{n+p} - u_n)$$

reduce, for long wavelengths, to the continuum elastic wave equation

$$\frac{d^2 u}{dt^2} = v_s^2 \frac{d^2 u}{dx^2},$$

where u_n is the displacement of atom n and v_s is the velocity of sound.

Exercise 2 — *Linear chain with next nearest neighbour interactions*

[moderate]

Consider a linear chain of identical atoms at the positions $x_p = pa$, $p = 1, 2, \dots$. The interaction is harmonic. The coupling constant to the second nearest neighbours is ν times smaller than that of the nearest neighbours.

- (a) Find the dispersion relation $\omega(k)$.
- (b) For which integer values of ν does the maximum of the dispersion lie at wavenumbers $k < \pi/a$?
- (c) What is the maximum frequency of an undamped wave? Discuss especially the case $\nu = 2$.
- (d) Determine the speed of sound,

$$v_s = \left. \frac{d\omega}{dk} \right|_{k \rightarrow 0}.$$

Exercise 3 — *Atomic chain of two types of atoms*

[moderate]

Assume a linear chain of atoms with mass M_1 at the Bravais lattice points $x_n = na$ (lattice constant a), and atoms of mass $M_2 \neq M_1$ at positions $x_n + d$ with $d < a$. Assume only nearest neighbour coupling with coupling constant D .

- (a) Give the differential equations for the displacements $u_{1,n}$ and $u_{2,n}$ of the atoms in cell n .
- (b) Use the ansatz

$$\sqrt{M_\alpha} u_{\alpha,n} = A_\alpha e^{i(kna - \omega t)}, \quad \alpha = 1, 2,$$

and derive the dispersion relation for the optical and the acoustic branch. Sketch the first Brillouin zone.



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- (c) Draw the group velocity for both branches.
 - (d) Discuss the solutions for $ka \ll 1$ and for $ka = \pm\pi$. Compute the amplitudes A_1 and A_2 and sketch the motion.
 - (e) Discuss the case $M_1 \gg M_2$.
 - (f) Assume the basis consists of Na^+ and Cl^- ions and the nearest-neighbour coupling constant is 20 N/m. Compute the width $\Delta\omega$ of the optical branch and the energy gap between optical and acoustic branches.